

Qualitative methods and ethnography

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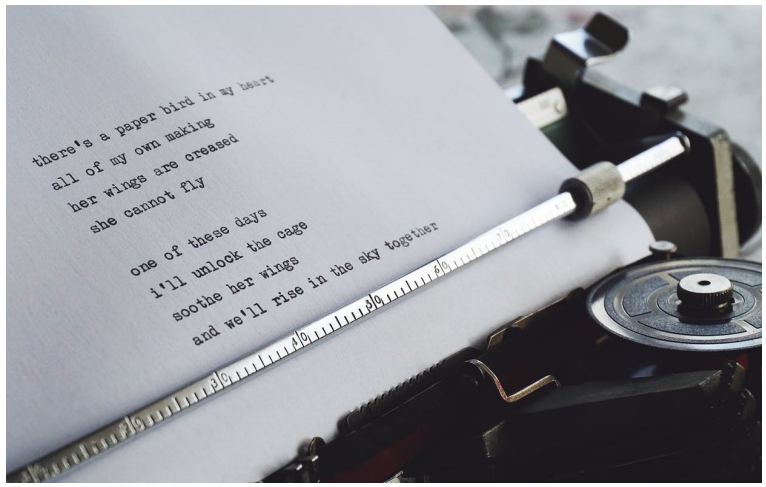
Content of the lecture

1) Theoretical principles of qualitative research:
   - Social constructionism
   - Phenomenology
2) Digital and design ethnography
3)Using ethnography in design process
4)Interviewing
5) Thematic analysis



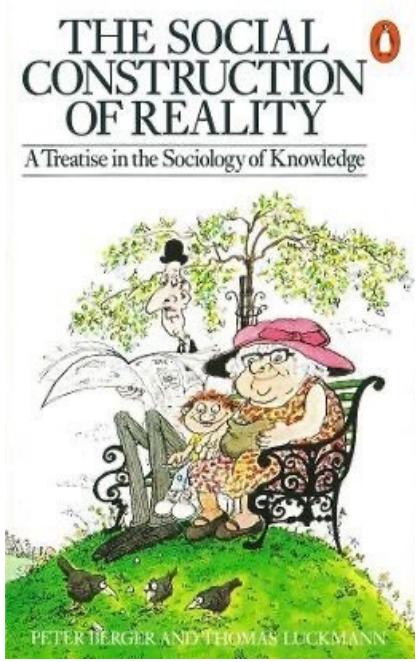
Principles of qualitative methods

- Focus on words and language
- Inductivist and constructionist logic
- Theory-driven approach
- *Interpretivism:* Understanding the world through an examination of the interpretation of that world by its participants
- *Constructionism:* Social properties are outcomes of the interactions between individuals, rather than phenomena 'out there' and separate from those involved in its construction.

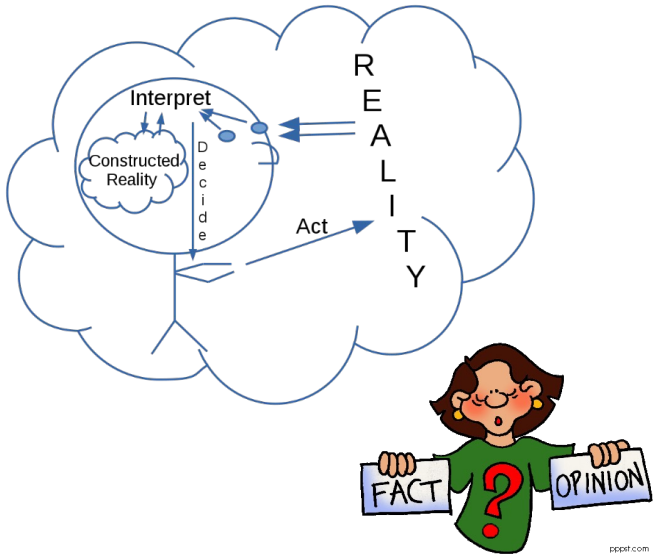


Social constructionism (Berger & Luckmann, 2002)

- Sociological theory of knowledge according to which human development is socially situated
- Knowledge is constructed through interaction with others
- Facts depend on the *shared ways of thinking* about and *representing the world* that groups of people develop collaboratively
- Institutionalization: When these roles are made available to other members of society, the reciprocal interactions are said to be institutionalized.
- Knowledge and people's conceptions (and beliefs) of what reality is become embedded in the institutional fabric of society.



Social constructionism



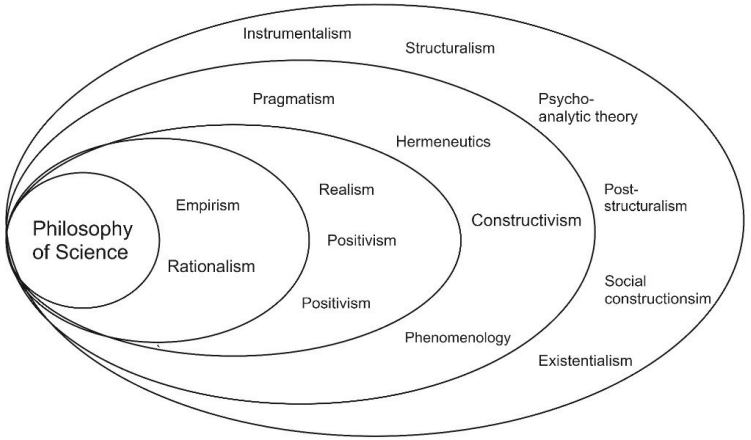
- Facts are constructed within language
- Each person and group create their own meaning of the reality
- Facts are representations, rather than physical entities
- Facts are institutionalised: embedded in social system, social network, organisation, and or society

- -- > STS-approach: Social construction of technology

Phenomenology (Husserl, Mearly-Ponty)

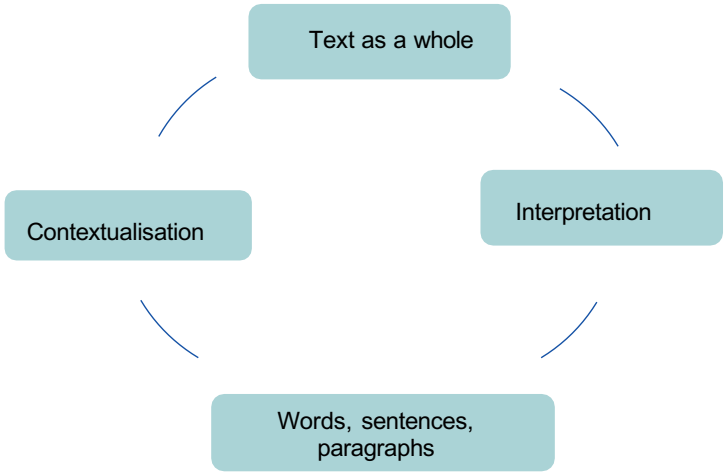
- Phenomenology (Edmund Husserl, Maurice Mearly-Ponty): Philosophical study of *experience* or *consciousness*
- Phenomenologists reject the concept of objective research
- Key principles:

1) Intentionality
2) Intuition
3) Intersubjectivity
4) Life-world



Hermeneutic circle (Heidegger, Gadamer)

- In-depth understanding of the phenomenon by interpretation, contextualisation and integration with theory
- Understanding text as a whole
- Individual parts (words, meanings) are interpreted to 'text as a whole'
- Meaning of a text must be found within its cultural, historical, and literary context (theory)
- Higher level of interpretation of data
- Inductive logic



Reflexivity in qualitative research

- Researchers should be reflective about the implications of their methods, values, biases, and decisions for the knowledge of the social world they generate
- Sensitivity to the researcher's cultural, political, and social context.

1) **Philosophical self-reflection:** an introspection involving 'an inward-looking, sometimes confessional and self-critical examination of one's own beliefs and assumptions'
2) **Methodological self-consciousness**: taking account of one's relationships with those whom one studies.
3) **Methodological self-criticism**: the confessional style of ethnographic writing



Positionality in qualitative research

- Researchers should be aware how their scholarly background and pre-knowledge could impact the conclusions drawn from the data
- Researchers' background shape both motivation and methodological choices

> *"We recognise that narratives portraying technology as inherently beneficial often obscure the risks of exclusion, overgeneralisation, and unintended harm, particularly for vulnerable populations. Rather than endorsing technology as a straightforward solution, we positioned ourselves as critical researchers attentive to power imbalances in design and deployment. This stance guided our commitment to participatory methods, ensuring that the lived experiences and design preferences of older adults were not marginalised but actively shaped the direction of the research."*

McCarren L, Erikson, U., Ortiz, L., Kuoppamäki S. (2026). Exploring the Design of a LLM- Based AI Assistant for Mindfulness Practice with Older Adults. (ACM Proceedings of the CHI Conference on Human Factors in Computing Systems)

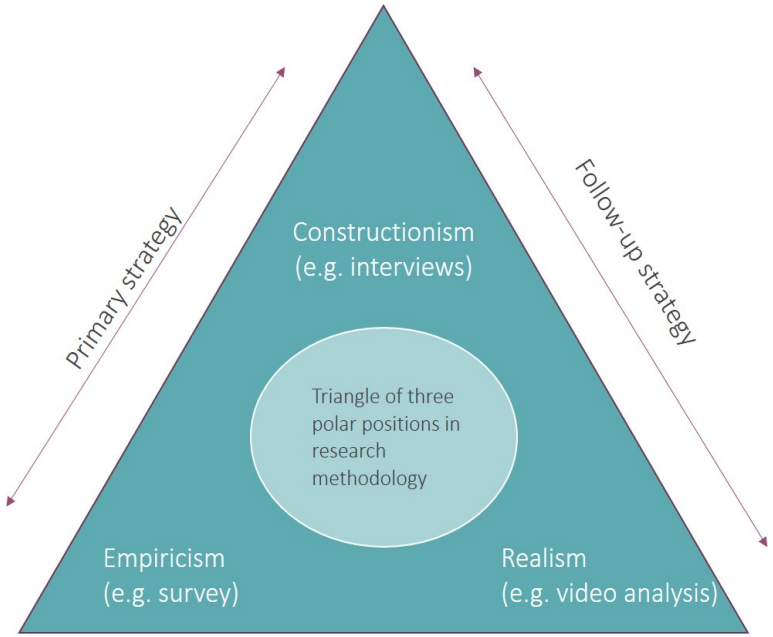
Reflection

- How does your (educational or personal) background impact the methodological choices, topics of your research or professional path?
- How does the values that are connected to certain fields or disciplines impact technical or other choices?
- Can you identify differences between disciplines?



Multiple triangulation (Olsen, 2004)

- Mixed method approach has become popular to answer both 'what' (quantitative) and 'how' and 'why' (qualitative) questions
- Primary strategy
- Follow-up strategy



Olsen, V. (2004). Triangulation in social research: Qualitative and quantitative methods can really be mixed. Developments in Sociology, ed. M. Holborn, Ormskirk: Causeway Press.

Ethnography and participant observation

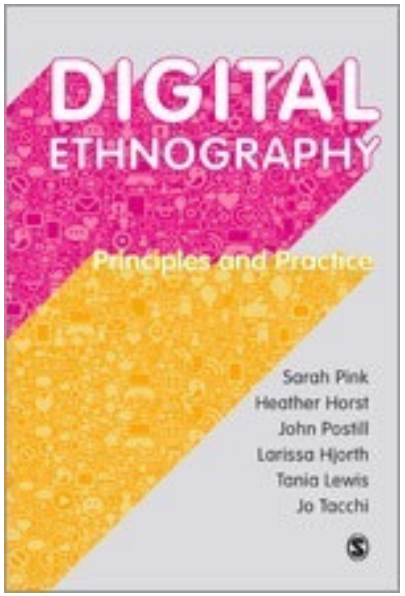
- Ethnographer immerses him- or herself in a social setting for an extended period of time
- Makes regular observations of the behaviour of members of that setting
- Listens to and engages in conversations;
- Interviews informants on issues that are not directly amenable to observation
- Collects documents about the group
- Develops an understanding of the culture of the group and people's behaviour within the context of that culture;
- Writes up a detailed account of that setting



Design and digital ethnography

- Using qualitative methods such as participant observation to improve technologies, products and services
- Design ethnographer informs the designers on human behaviour to make better products or services
- Principles of digital ethnography (Pink et al., 2016)

1) Multiplicity
2) Openness
3) Reflexivity



Ethnography in HRI

- Using videos (videography) to capture the complexity of robot interactions in real-life environments
- Setting 1: Fully controlled lab environment (e.g. robot in the laboratory)
- Setting 2: Not fully controlled settings (e.g. robot in the public space)
- Setting 3: Institutional real-world settings (e.g. robot ”in the wild”)

Fig. 10: Total view of the elderly care home common room.



Arne Maibaum, Philipp Graf, René Tuma (2021) On the Use of Videography in HRI

Video in Anthropological Research

1)Video (walking with video)

2)Video in homes

3)Other people's videos



https://youtu.be/k-BW1piP6Is

Ethnography in design research

- Group discussion 1 (10 min)

- What are the benefits of using anthropological methods in biomedical engineering education? In what ways ethnography or anthropological methods contributed to the product design?

- What are different methods presented in the articles? How would you apply these methods in your project?

- **Google Doc:** https://bit.ly/3WJhlBh



Break (5 min)



Interviewing: Basic principles

- Interest in the *interviewee's point of view*; (vs. quant. research that reflects the pre-defined concerns)
- 'Rambling' or going off topic is often encouraged - it gives insight into what the interviewee sees as relevant
- Interviewers can depart significantly from any schedule or guide that is being used. They can ask new questions that follow up interviewees' replies and can vary the order and even the wording of questions.

- Qualitative interviewing is flexible, responding to the direction in which interviewees take the interview
- Researcher wants to have rich, detailed answers (vs. quant research that aims to generate answers that can be coded quickly)
- Interviewee may be interviewed more than once - sometimes several times.

Semi-structured interview

- Interview guide: the researcher has a list of questions or fairly specific topics
  Flexibility is important! Questions may not follow on exactly in the way outlined on the schedule
- Some level of structure, but allowing interviewees to pursue topics they find interesting

1) Formulate interview questions in a way that will help you to answer your research questions (but try not to make them too specific);
2) Try to use a language that is comprehensible and relevant to the people you are interviewing;
3) Do not ask leading questions;
4) Remember to ask background information if relevant to your research topic

Remember: Interview is a gift!

- Trust
- Empathy
- Respect



Elicitation Techniques: Getting People to Talk About Ideas They Don't Usually Talk About

- Social science researchers often ask about topics that are difficult to discuss in formal interviews (e.g. controversial topics, abstract concepts, religious, gender or sexual identities, ideas that may be tacit and not explicitly known)
- Use visual, verbal, or written stimuli to encourage participants to talk about their ideas
- Sorting
- Ordering
- Drawing
- Think aloud
- Photo-eliciation



Figure 1. Image for Eliciting Students' Understanding of the Perspective of an Escaping Slave (Still, 1872, p. 312; public domain)

"Imagine what the characters are thinking? Is that what she would think or what you would be thinking?" or "What about if you went back in time, what would you be thinking?"

"Tacit knowledge"

- *Tacit knowledge:* Knowledge that is seldom expressed that guides professional practice. Personal experience, intuition, insight. (vs. formal or explicit knowledge)
- *Eliciation technique*: Using videos or photography as a part of interview to uncover 'tacit knowledge'
- Videos, photography or other material will 'elicit' information that would otherwise remain below the surface. They provide a concrete reference point which that interviewees can recognise, reflect and elaborate upon.



- van Braak, M., de Groot, E., Veen, M., & Giroldi, E. (2018). Eliciting tacit knowledge: The potential of a reflective approach to video-stimulated interviewing. *Perspectives on medical education*, *7*(6), 386–393. https://doi.org/10.1007/s40037-018-0487-9

Pre-interview

1. Introductory questions
2. Physical activity – habits and motivations
3. Perceived health and well-being
4. Use of technologies for physical activity
   - If the person has / has not used any fitness applications
5. Digital skills
6. Use of wearables for physical activity
   - If the person has / has not used any wearables
7. Physical activity and social needs

Post -interview

- Experiences with wearables with real-time feedback
- Feedback, Data, and Interaction
- Emotional and Physical Responses
- System usability and future use

Lu, T-Y., Rosato, A., Dual, S., Kuoppamäki, S. (2026). Lived experiences of older adults using wearables with real-time feedback: Phenomenological study. *JMIR mHealth and uHealth*. doi: 10.2196/71509

Post-interview questions (examples)

2. Feedback, data and interaction	Can you share a specific occasion when you used WRF and how you experienced it? What was the main reason you started using a wearable device? Why did you begin using it? What was your experience with the wearable device? Do you think it helped you with your physical activity? Do you think the device helped you become more motivated? Why or why not? Have you noticed any other benefits of using the device? Did you face any challenges, disadvantages, or limitations when using the device? Was it easy to start using the wearable device?

Post-interview questions (examples)

1. Experience with the WRF	What do you think about wearing the device on your chest? Did you receive any feedback (health-related information) from the wearable device? If so, what type of feedback did you find useful or less useful? Do you like the idea of receiving feedback from the device? What type of feedback from the device would motivate you to exercise more? Could you imagine having a conversation with a wearable device?

Thematic analysis

1) Defining research questions
2) Reading the text as a whole
3) Selecting the unit of analysis (words, sentences, paragraphs)
4) Coding sub-cateogories
5) Reading text again
6) Creating generic categories (based on sub-categories)
7) Abstraction: creating main categories

Elo, S., Kyngäs, H. (2008). The qualitative content analysis process. *Journal of Advanced Nursing*. https://doi.org/10.1111/j.1365-2648.2007.04569.x

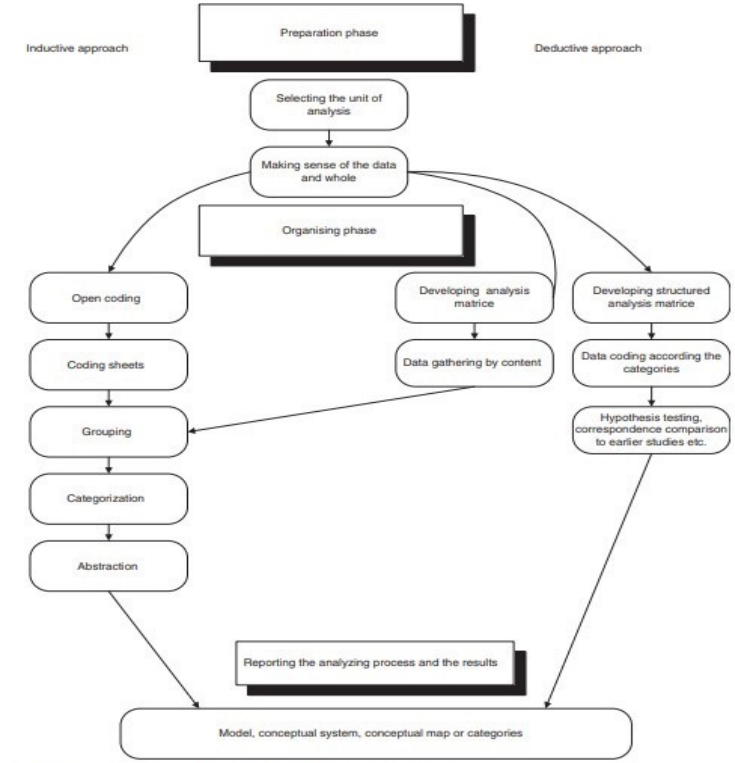


Figure 1 Preparation, organizing and resulting phases in the content analysis process.

Example of transcription

P6: Jag tycker ibland att det är väldigt svårt när man ska ta ett recept på..på sin, padda, och det åker bara bort hela tiden..då vore det ju bra om roboten hade det inne i sig, liksom till att diskutera receptet med med, det tycker jag skulle vara ganska skönt..

P3: Det var precis det jag tyckt men en liten bit till, och alla andra appar som man också måste ha och sådär så man liksom blir van vi det..

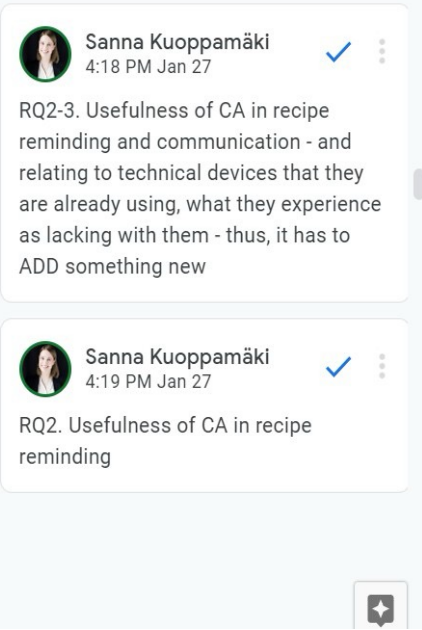
S. Mm men är det liksom mer att diskutera, och inte som mer att liksom..förslag eller..

P6: ..nej, nej men..vi följer receptet tillsammans, och sen, nu säger nu tar du det här och det här, istället för att jag ska springa tillbaka till datorn och trycka fram receptet igen och hålla på, då kunde dat..roboten ha i si..inprogrammerat, och jag slipper hålla på och leta och leta på min padda då..

P1: Och vi ska bara fråga då, 'vad ska vi göra nu?'

P6: ..ja precis…precis..

P7: ..och sen säger den helt plötsligt 'du har glömt att köpa socker'..



Social alarm system in nursing homes

- **Data extraction:**

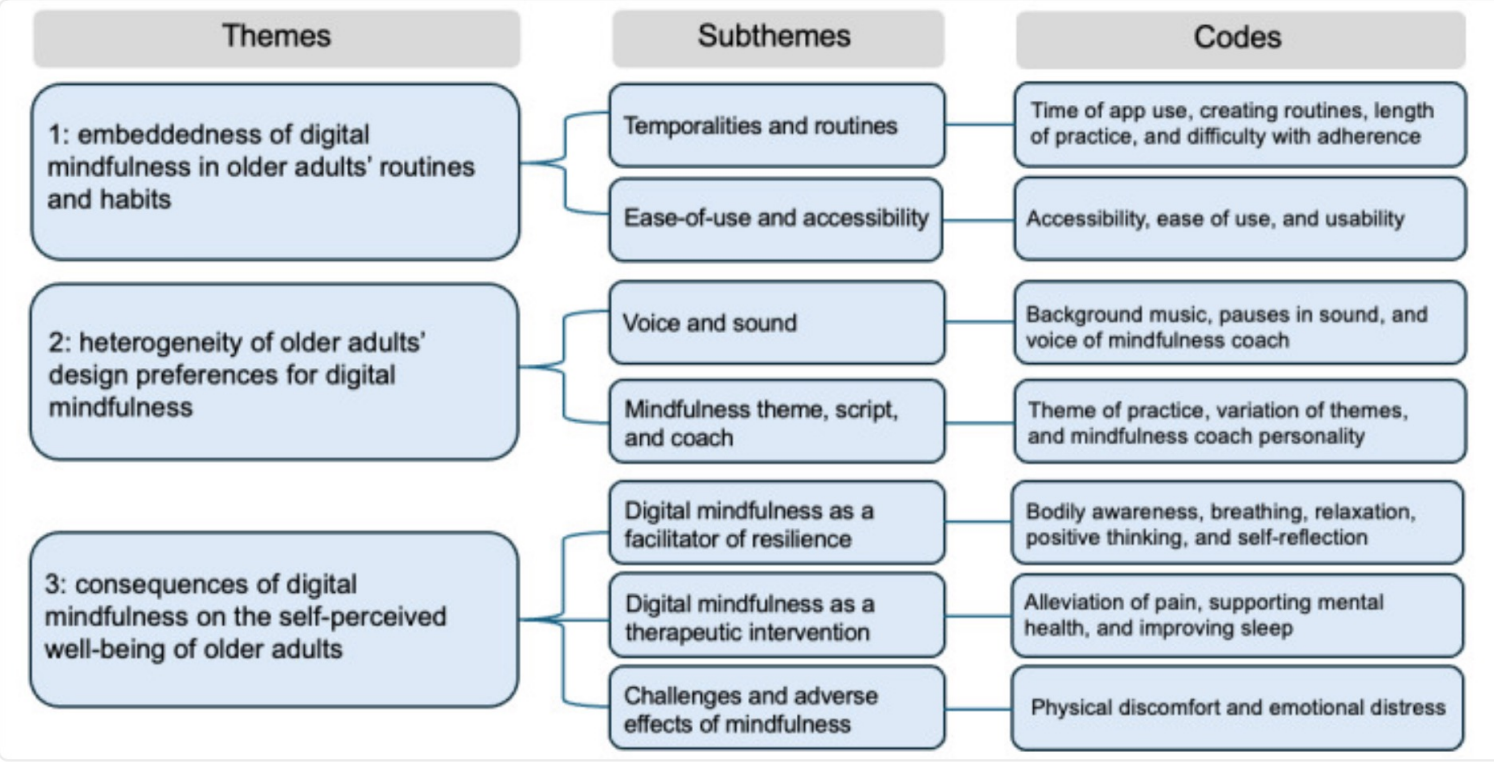
'As you can see, we receive tones of alarms... The system helps us to get closer to our patients... But it is challenging to deal with all the requests within a limited time. They send you alarms, they expect your reply promptly… This negotiation requires energy and time, the unlimited alarms kind of make me more stressed than before.'

- (Assistant nurse)

What are the benefits and barriers of using alarm system in care delivery in nursing homes?

Data extraction	Sub-category	Generic categor y	Main category
”Alarm system improves the care delivery”	Care delivery	Effectiveness	Benefits of alarm system usage in health care
”Nurses use alarm systems in care provision”	Devices and systems	Accessibility	
”Alarm systems are used for communication with the patient”	Patient-relationship	Quality of care	

User experience of digital mindfulness



More examples from thematic analysis

- Zhang, X., Zayed, A., Hamrin, J. Guneysu, A., Kuoppamäki, S. (2025). Exploring Body Image Awareness With a Large Language Model–Based Conversational Agent: Qualitative Study With Young Adults. *Journal of Medical Internet Research*. doi: 10.2196/78829
- Lu, T-Y., Rosato, A., Dual, S., Kuoppamäki, S. (2026). Lived experiences of older adults using wearables with real-time feedback: Phenomenological study. *JMIR mHealth and uHealth*. doi: 10.2196/71509
- McCarren, L., Eriksson, U., Mengual, L., Kuoppamäki, S. (2026). Exploring the Design of an LLM-Based AI Assistant for Mindfulness Practice with Older Adults. In *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems (CHI '26)*. Association for Computing Machinery, New York, NY, USA, Article 253, 1–16. https://doi.org/10.1145/3772318.3790734
- McCarren, L., Kuoppamäki, S. (2025). Design Preferences, Routines, and Well-Being of Older Adults Using Voice-Guided Digital Mindfulness: Qualitative Interview Study**.** *JMIR Human Factors*, 12:e67533. DOI: 10.2196/67533
- Lu, T-Y., Dual, S., Kuoppamäki, S. (2026). Moving Together: A Co-Design Study on Wearable Materials and Aesthetics for Active Ageing. *ACM Designing Interactive Systems (DIS)* (Accepted for publication)

What's next?

- Your user study should include a selection and description of method to study users + plan for data collection
- Reflection in small groups (10 min):

  - Who would you interview for your user study?
  - What questions to ask?
  - What themes to include?
  - How to get access to 'tacit knowledge'?



- Use Padlet: https://padlet.com/sannaku/incpxmhibl4igr0k

- Answer the same questions in your short reaction paper!